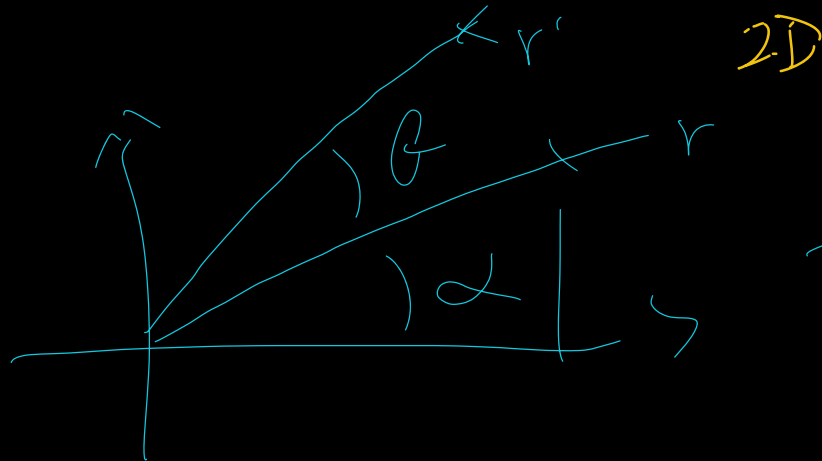




2D

$$\begin{cases} x = |r| \cos \alpha \\ y = |r| \sin \alpha \end{cases}$$

$$\begin{cases} x' = |r| \cos (\alpha + \beta) \\ y' = |r| \sin (\alpha + \beta) \end{cases}$$

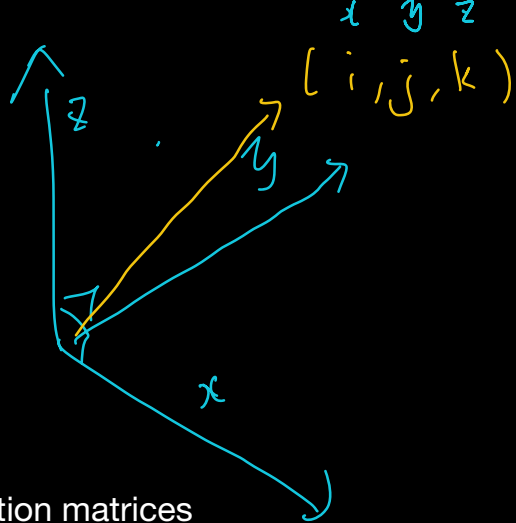
2d rotation matrix

$$\begin{aligned} x' &= |r| (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ &= x \cos \beta - y \sin \beta \end{aligned}$$

$$\begin{aligned} y' &= |r| (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ &= x \sin \beta + y \cos \beta \end{aligned}$$

∴ The rotational matrix with an angle θ on a 2-D plane.

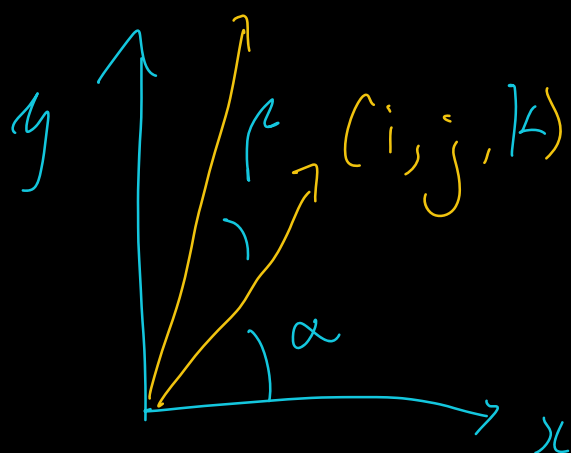
$$\therefore \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



3d rotation matrices

Anti-clockwise.
rotation along the z -axis,
i.e., on the x - y plane.

$$\begin{cases} x = |r_z| (\cos \alpha) \\ y = |r_z| (\sin \alpha) \\ z = k \end{cases}$$

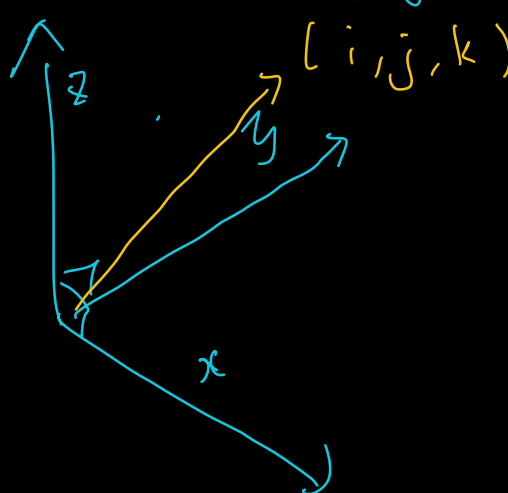


$$\begin{cases} x' = |r_z| (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ y' = |r_z| (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ z = k \end{cases}$$

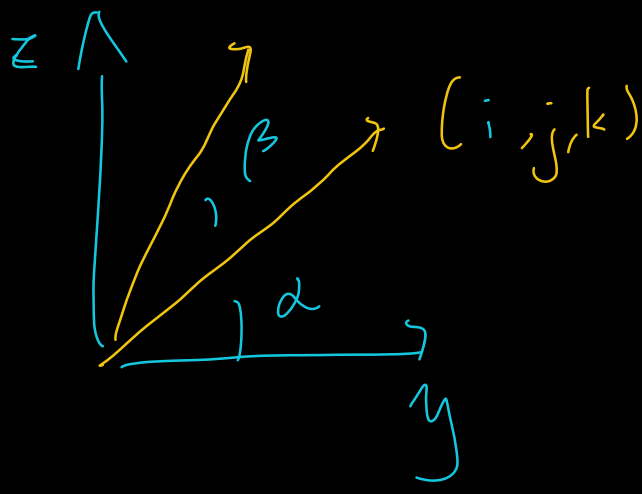
$$\therefore \begin{cases} x' = x \cos \beta - y \sin \beta + z (0) \\ y' = x \sin \beta + y \cos \beta + z (0) \\ z' = z (1) + z (0) + z (0) \end{cases}$$

$\therefore$ the rotational matrix around z axis

$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$


 Anti clock Rotation along x axis.

$$\begin{cases}
 x = i \\
 y = |r_x| \cos \alpha \\
 z = |r_x| \sin \alpha
 \end{cases}$$



$$\begin{cases}
 x' = i \\
 y' = |r_x| (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\
 z' = |r_x| (\sin \alpha \cos \beta + \cos \alpha \sin \beta)
 \end{cases}$$

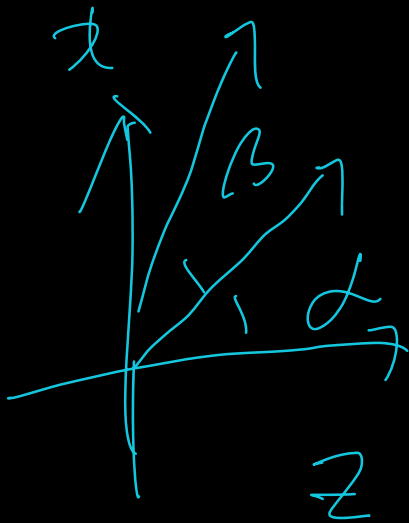
$$\therefore \begin{cases}
 x' = x \\
 y' = y \cos \beta - z \sin \beta \\
 z' = y \sin \beta + z \cos \beta
 \end{cases}$$

$$\therefore \begin{bmatrix}
 1 & 0 & 0 \\
 0 & \cos \theta & -\sin \theta \\
 0 & \sin \theta & \cos \theta
 \end{bmatrix}$$

auto-dock. Along y axis.

$$\begin{cases} x = |r_y| \sin \alpha \\ y = j \\ z = |r_y| \cos \alpha. \end{cases}$$

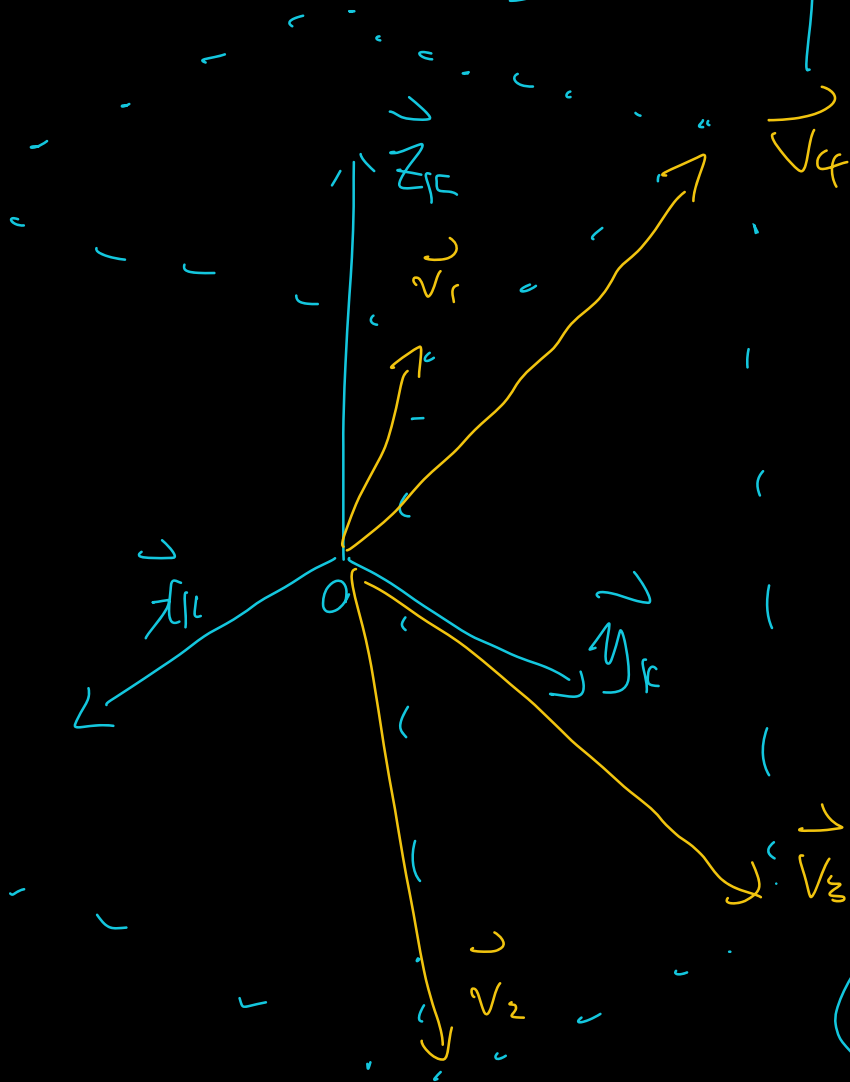
$$\begin{cases} x' = |r_y| (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ y' = j \\ z' = |r_y| (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \end{cases}$$



$$\begin{cases} x' = x \cos \beta + z \sin \beta \\ y' = y \\ z' = -x \sin \beta + z \cos \beta. \end{cases}$$

$$\therefore \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

given vectors from $(0, 0, 0)$ to faces, where
vectors $\perp$ corresponding faces.

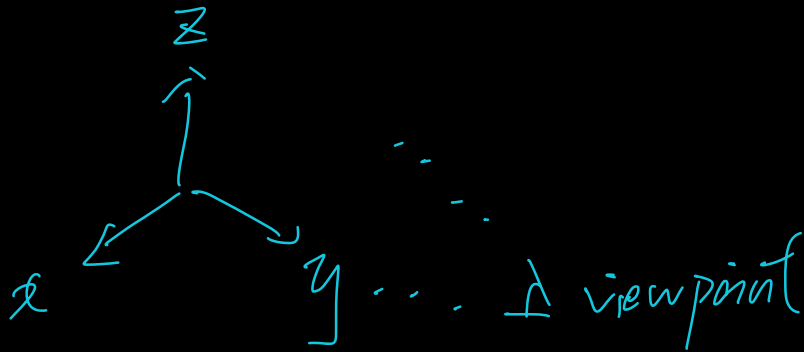


to find the vectors
to the vertices of the
cube face towards the
+y direction:

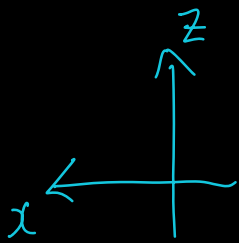
$$\begin{cases} \vec{v}_1 = \vec{y}_k + \vec{x}_k + \vec{z}_k \\ \vec{v}_2 = \vec{y}_k + \vec{x}_k - \vec{z}_k \\ \vec{v}_3 = \vec{y}_k - \vec{x}_k - \vec{z}_k \\ \vec{v}_4 = \vec{y}_k - \vec{x}_k + \vec{z}_k \end{cases}$$

Determine whether a point is within the 2d projection of the shown faces

1. $3D \rightarrow 2D$.



i.e., dropping the y dimension and we get. xz plane.
(coordinates)



2. To determine whether a point is within the 2D-projection, we need:

1. vectors of the edges.
2. vectors of the point with the same starting point as the edges.

3. The cross product between 1 & 2.

★ using cross product only works when the shape is convex.

2.1 vectors of edges:

$$\left\{ \begin{array}{l} \vec{v}_2 - \vec{v}_1 \\ \vec{v}_3 - \vec{v}_2 \\ \vec{v}_4 - \vec{v}_3 \\ \vec{v}_1 - \vec{v}_4 \end{array} \right.$$

2.2 vectors of the point.

$$\left\{ \begin{array}{l} \vec{u} - \vec{v}_1 \\ \vec{u} - \vec{v}_2 \\ \vec{u} - \vec{v}_3 \\ \vec{u} - \vec{v}_4 \end{array} \right.$$

2.3. Cross Product

$$(\vec{v}_{i+1} - \vec{v}_i) \times (\vec{u} - \vec{v}_i)$$

if the point is within the 2D projection of a face,
the cross products should stay CONSISTENTLY
positive or negative.

(to the left) (to the right)